Ensemble Learning for Robust Image Classification under Distribution Shift

Stephanus Park
Otto-Friedrich University of Bamberg
96049 Bamberg, Germany
stephanus.park@stud.uni-bamberg.de

xAI-Proj-M: Domain Generalization for Robust Machine Learning
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Sebastian Drefahl
Otto-Friedrich University of Bamberg
96049 Bamberg, Germany
sebastian.drefahl@stud.uni-bamberg.de

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Abstract

Robustness under distribution shift remains a central challenge in computer vision. While ensemble methods are widely used to improve generalization, their effectiveness depends on both model diversity and aggregation strategy. In this work, we compare homogeneous and heterogeneous ensembles under synthetic and real-world distribution shifts, and evaluate whether adaptive expert selection with a mixture-of-experts routing mechanism can better exploit model diversity

We construct a homogeneous CNN ensemble and a heterogeneous ensemble combining convolutional networks with a Vision Transformer. Performance is evaluated on clean data, synthetic corruption benchmarks, and a student-collected out-of-distribution dataset. Oracle analysis is used to quantify the potential benefit of complementary expertise.

Results show that homogeneous ensembles provide limited robustness gains due to correlated failures under shift. In contrast, heterogeneous ensembles exhibit larger oracle gaps and improved robustness in severe corruption regimes. However, a learned mixture-of-experts router primarily collapses to a dominant expert, yielding only modest improvements over uniform averaging. These findings indicate that ensemble robustness is constrained both by expert diversity and by the ability of routing mechanisms to effectively leverage that diversity.

1 Introduction

Deep learning models for image classification are usually developed under the assumption that training and test data follow the same distribution, yet real deployments routinely violate this condition. Changes in acquisition device, environment, or image quality induce distribution shifts that degrade performance, even for models that achieve high in-distribution accuracy (Quiñonero-Candela et al., 2008). This gap between benchmark performance and real-world reliability is a core challenge in domain generalization and out-of-distribution (OOD) robustness.

Improving clean accuracy alone does not guarantee robustness under domain shift, and modern vision models can fail abruptly on corrupted or unseen inputs while remaining highly confident in their predictions (Zhou et al., 2023). Such overconfident failures limit the applicability of deep models in safety- or reliability-critical applications and motivate methods that explicitly target robustness and calibration rather than only in-distribution accuracy.

Ensemble learning offers a practical way to improve robustness by combining models that fail in different ways under distribution shift (Dietterich, 2000). Heterogeneous ensembles that mix architectures tend to respond differently to the same domain shift than ensembles consisting of multiple instances of the same backbone (Petrakova et al., 2015). By aggregating predictions from such diverse experts, ensembles can achieve better robustness than single models or homogeneous ensembles (Wu et al., 2023).

Beyond static ensembling, mixture-of-experts (MoE) models introduce a routing mechanism that assigns inputs to experts in an input-dependent way (Masoudnia and Ebrahimpour, 2014). Rather than averaging all experts uniformly, a learned router can upweight experts that are better suited to a given input, potentially exploiting architectural diversity more efficiently under distribution shift. Yet it remains unclear whether routers trained on realistic data naturally specialize experts or instead collapse to using a single dominant model, leaving potential robustness gains untapped (Fedus et al., 2021).

In this work, we empirically study ensemble robustness under distribution shift with a focus on architectural diversity and expert routing. Concretely, we compare homogeneous ensembles of ResNet models with heterogeneous ensembles that combine multiple CNN backbones and a ViT, and we evaluate these ensembles on a subset of the ImageNet dataset under synthetic corruptions and on a custom phone-captured test set to quantify robustness under realistic distribution shifts. Our main contributions are: (1) a controlled comparison of homogeneous and heterogeneous ensembles under synthetic and real-world distribution shifts; (2) evidence that architectural heterogeneity improves robustness under domain shift, even when gains in clean accuracy are limited; and (3) an analysis of MoE routing behavior showing that, without explicit incentives for specialization, routing tends to over-rely on a single expert despite the presence of complementary models.

2 Background

2.1 Distribution Shift

In realen Anwendungsszenarien unterscheiden sich Trainings- und Testsituationen häufig in relevanten Eigenschaften. Diese Abweichungen werden im Rahmen des Dataset-Shift-Problems beschrieben, das sich mit der Übertragung von Information zwischen eng verwandten, aber nicht identischen Umgebungen befasst (Quiñonero-Candela et al., 2008, pp. 4–5). Wie verläßlich sind dabei Vorhersagen mit veränderten Testdaten, die nicht aus der Trainingsumgebung stammen, mit denen das Model trainiert wurde?

Klassische überwachte Lernverfahren sind in der Regel nicht darauf ausgelegt, mit solchen Distribution Shifts umzugehen. Die Vorhersageleistung werden unter der impliziten Annahme stabiler Trainings- und Testbedingungen optimiert und können daher bereits bei moderaten Abweichungen der Datenverteilung deutliche Leistungseinbußen zeigen (Quiñonero-Candela et al., 2008, p. 28), Zhou et al. (2023). Diese Beobachtung verdeutlicht, dass hohe Genauigkeit unter Trainingsbedingungen allein kein verlässlicher Indikator für Robustheit in realen Anwendungsszenarien ist und motiviert Ansätze, die explizit auf Generalisierung unter Distribution Shift abzielen.

2.2 Domain Generalisierung *(DG)* and Robustness *(RB)*

Viele trainierte Modelle zeigen eine hohe Leistung auf dem jeweiligen Trainingsdatensatz, generalisieren jedoch nur unzureichend auf Daten aus anderen Erhebungsumgebungen. (Torralba and Efros, 2011) zeigen in einer empirischen Analyse, dass gängige Bilddatensätze starke, datensatzspezifische Biases enthalten, die zu erheblichen Leistungseinbrüchen bei Cross-Dataset-Evaluationen führen.

Modelle lernen dabei nicht nur aufgabenrelevante Merkmale, sondern auch domänenspezifische Regularitäten, die außerhalb der Trainingsdomäne nicht stabil sind.

Domain Generalization adressiert dieses Problem, indem das Lernen unter expliziter Berücksichtigung von Verteilungsverschiebungen zwischen Domänen betrachtet wird. Im Gegensatz zur Domain Adaptation wird dabei angenommen, dass während des Trainings kein Zugriff auf Daten aus der Ziel-Domäne besteht (Muandet et al., 2013). Ziel ist es, Repräsentationen zu lernen, die über mehrere Trainingsdomänen hinweg stabil sind und auf zuvor unbekannte Domänen übertragen werden können.

Formal liegt der Domain-Generalization-Annahme zugrunde, dass sich die Eingabeverteilungen zwischen Domänen unterscheiden, also $P_k(X) \neq P_l(X)$, während die bedingte Verteilung der Zielvariable gegeben die Eingabe über Domänen hinweg stabil bleibt, d. h. $P_k(Y \mid X) = P_l(Y \mid X)$, für unterschiedliche Domänen $\mathcal{D}_k$ und $\mathcal{D}_l$ Muandet et al. (2013).

(Muandet et al., 2013) formulieren Domain Generalization als das Problem, aus mehreren verwandten Domänen solche Merkmale zu extrahieren, die invariant gegenüber domänenspezifischen Veränderungen sind, während die zugrunde liegende funktionale Beziehung zwischen Eingabe und Zielvariable erhalten bleibt. Damit stellt DG einen zentralen Ansatz dar, um Robustheit gegenüber systematischen Verteilungsverschiebungen zu erreichen, ohne auf nachträgliche Anpassung an neue Domänen angewiesen zu sein.

2.3 Ensemble Learning *EL*

Da DG und RB darauf abzielen, Leistungsabfälle unter unbekannten Verteilungsverschiebungen zu vermeiden, stellt sich die Frage nach geeigneten Ideen, zur Erhöhung der Robustheit. Im survey von Zhou et al. (2023) wird EL als modellbasierter Ansatz zur DG eingeordnet, bei dem Robustheit durch die Kombination mehrerer Modelle oder Modellzustände erzeugt wird. Die Idee dabei ist, dass unterschiedliche Modelle unterschiedlich auf Distribution Shifts reagieren und somit nicht auf denselben Eingaben versagen. Durch die Aggregation mehrerer Hypothesen können solche komplementären Fehlermuster ausgenutzt und die Abhängigkeit von einem einzelnen Modell reduziert werden.

Die Motivation besteht darin, dass unterschiedliche Modelle unterschiedliche Fehlermuster aufweisen und somit nicht auf denselben Eingaben scheitern. Durch die Aggregation mehrerer Modelle können solche komplementären Fehler ausgenutzt und die Varianz einzelner Vorhersagen reduziert werden.

Im Kontext von DG und RB gewinnt EL besondere Bedeutung. Unter Verteilungsverschiebungen reagieren Modelle oft unterschiedlich auf neue oder veränderte Daten, sodass kein einzelnes Modell über alle Domänen hinweg optimal ist. Ensembles bieten hier eine Möglichkeit, Robustheit auf Modellebene zu erhöhen, indem mehrere Hypothesen berücksichtigt und ihre Vorhersagen kombiniert werden.

Mit diesem Wissen und den vorhandenen Ressourcen wurde die Ensemble Methode als geeigneter Projektansatz gewählt. Dabei bot sich im Zusammenhang mit den zuvor trainierten 18-layer und 50-layer ResNets an, ein homogenes Ensemble zu untersuchen und dem gegenüber ein heterogeneren Ensembleansatz zu stellen, entlang der Diversität in den Modellen.

3 Methodology

3.1 Setup

The experiments were conducted using multiple computational setups. Training was performed on several Windows-based workstations equipped with NVIDIA A4000 GPUs, using appropriately configured development environments managed with `conda`. In addition, training runs were executed on Google Colab Pro, utilizing NVIDIA A100 and T4 GPUs. To ensure availability and persistence of data, results, and trained models, a pipeline integrating GitHub and Google Drive was established. For each training run, the two best-performing models as well as the final trained model were stored on Google Drive. Experiments were organized using Git branches, and the corresponding notebooks were retrieved from GitHub, modified when necessary, and versioned accordingly.

All training runs and evaluations were logged using the online experiment tracking service *Weights & Biases (WB)*[1]. The platform enabled rapid inspection of intermediate results and provided aggregated metrics to support further optimization.

We implemented several training setups. Initially, the goal was to train two models from the residual nets, namely 18-layer and 50-layer ResNet. These models served as an initial baseline architectures for subsequent development and comparison. During training, hyperparameter sweeps were conducted to identify the optimal learning rate and the corresponding batch size. The primary focus was on achieving a high *validation accuracy*. The resulting models were retained and reused in later stages of the project.

At this stage, the main objective was to become familiar with the training setup and to establish a functional workflow covering data transfer, model training with validation, and systematic evaluation of results.

3.2 Homogeneous Ensemble

Der Aufbau eines Ensembles aus mehreren ResNet-Varianten ist ein methodisch sinnvoller Ansatz. ResNets mit unterschiedlicher Tiefe, wie das 18-layer, 34-layer und 50-layer ResNet, unterscheiden sich in ihrer Modellkapazität, in der Tiefe ihrer Repräsentationen sowie in ihrem Optimierungsverhalten. Obwohl alle Modelle auf derselben Grundarchitektur basieren, zeigen He et al. (2016), dass innerhalb der ResNet-Familie eine relevante Diversität vorhanden ist.

Diese Diversität ergibt sich aus der unterschiedlichen Anzahl an Residual-Blöcken. Durch diese Blöcke werden hierarchische Verfeinerungen der Merkmalsrepräsentationen ermöglicht. Tiefere Modelle, insbesondere das 50-layer ResNet, können dadurch komplexere und abstraktere Repräsentationen lernen. Flachere Modelle wie das 18-layer ResNet lernen hingegen andere Aktivierungsmuster und sind stärker durch lokalere Transformationen geprägt. In der Folge besitzen die Modelle unterschiedliche induktive Biases und gewichten verschiedene Aspekte derselben Eingabe unterschiedlich.

He et al. (2016) zeigen außerdem, dass ResNets unterschiedlicher Tiefe nicht nur unterschiedliche Genauigkeiten erreichen, sondern auch unterschiedliche Fehlercharakteristika aufweisen. Diese Fehler treten nicht bei denselben Eingaben auf, sondern sind teilweise komplementär. Für den Einsatz eines Ensembles ist dies entscheidend, da sich Fehlentscheidungen einzelner Modelle durch korrekte Vorhersagen anderer Modelle ausgleichen können. Die hohe Leistungsfähigkeit der Einzelmodelle kann dadurch im Ensemble besser ausgenutzt werden als im Einsatz eines einzelnen tiefen Modells.

Da alle Modelle derselben Architekturklasse angehören, unter denselben Trainingsbedingungen und auf demselben Datensatz trainiert werden, bleibt das Ensemble homogen und gut interpretierbar. Ein ResNet-Ensemble eignet sich daher besonders gut als Baseline, um Ensemble-Effekte, Modellvielfalt und Robustheit systematisch zu untersuchen.

Zudem lässt sich ein ResNet-Ensemble gut mit einem einzelnen 50-layer ResNet vergleichen. Da das 50-layer ResNet das leistungsstärkste Einzelmodell innerhalb des Ensembles darstellt, ermöglicht dieser Vergleich eine klare Einordnung des Ensemble-Effekts. Leistungsgewinne des Ensembles lassen sich so direkt auf die Kombination mehrerer Modelle zurückführen und nicht allein auf eine erhöhte Modellkapazität. Dadurch wird eine saubere Abgrenzung zwischen Verbesserungen durch Ensembling und solchen ermöglicht, die lediglich aus dem Einsatz eines starken Einzelmodells resultieren.

Aus Sicht des Ensemble Learnings ist die Kombination mehrerer Modelle mit unterschiedlichen induktiven Biases besonders geeignet, um Varianz zu reduzieren und die Robustheit zu erhöhen. Flachere ResNet-Varianten tendieren dazu, stärker lokale und niedrigstufige visuelle Muster zu nutzen, während tiefere ResNets abstraktere und semantisch reichhaltigere Repräsentationen lernen. Ein Ensemble aus ResNet-Modellen unterschiedlicher Tiefe ermöglicht es daher, kontrolliert Vorhersagediversität zu erzeugen, ohne die zugrunde liegende Architekturklasse zu verändern. Dies macht ResNet-basierte Ensembles besonders geeignet als Ausgangsbasis für die Analyse von Robustheit und Fehlervielfalt unter Verteilungsverschiebungen.

[1] Weights & Biases. (n.d.). *Weights & Biases: Machine learning experiment tracking*. https://wandb.ai/site (accessed February 10, 2026).

3.3 Heterogeneous Ensemble

To test the effect of architectural diversity, we construct a heterogeneous ensemble from four backbones with distinct biases, all fine-tuned on the same ImageNet-subset training data.

These backbones are summarized in Table 1. They were specifically chosen to maximize inductive bias diversity: the CNNs provide complementary local feature extraction (texture, details, scaling), while the Vision Transformer (ViT) adds global reasoning. Identical training ensures differences arise from architecture alone.

Table 1: Backbone architectures comprising the heterogeneous ensemble. Each model is initialized with ImageNet-pretrained weights and fine-tuned on the same training data to isolate architectural inductive bias effects.

Backbone	Key characteristics	Reference
ResNet-34	strong local texture bias, stable baseline	He et al. (2016)
DenseNet-121	Dense connectivity for feature reuse	Huang et al. (2017)
EfficientNet-B0	Parameter-efficient, balanced scaling	Tan and Le (2019)
ViT-B/16	Global self-attention, shape-focused	Dosovitskiy et al. (2020)

Predictions are aggregated via simple averaging:

$$\hat{y} = \frac{1}{K} \sum_{k=1}^{K} p_k(\mathbf{x})$$

where $p_k(\mathbf{x})$ is the softmax output of expert k, and $K = 4$. This uniform weighting provides a baseline before introducing learned routing in Section 3.4.

Prior work suggests that architectural diversity can induce complementary failure modes under distribution shift (Wu et al., 2023). We evaluate this hypothesis empirically.

3.4 Mixture-of-Experts Router

To adaptively exploit architectural diversity, we extend the heterogeneous ensemble with a mixture-of-experts (MoE) router (Masoudnia and Ebrahimpour, 2014). Rather than uniformly averaging predictions, the router produces input-dependent weights over experts.

The routing network is a lightweight two-layer MLP that takes penultimate-layer feature representations from the ViT-B/16 backbone as input and outputs a probability distribution over experts via softmax. ViT features are used as routing input due to their global, content-aware representations, which are well-suited for identifying image characteristics associated with different failure modes.

Given gating weights $\mathbf{g} \in \Delta^3$, the final prediction is computed as

$$\hat{y} = \sum_{k} g_k p_k(\mathbf{x}).$$

The router is trained using cross-entropy loss on an 80/20 split of validation set, while all expert backbones remain frozen. Strong regularization, including heavy dropout, is applied to reduce the risk of expert collapse, as observed in large-scale MoE systems (Fedus et al., 2021).

This setup allows us to directly study whether expert diversity alone is sufficient for effective routing, or whether additional incentives for specialization are required.

4 Experimental Setup

4.1 Datasets

Experiments are conducted on a 10-class object classification task provided within the xAI-Proj-M framework. Standard evaluation is applied on the clean set, while robustness is evaluated under two types of distribution shifts.

Clean Dataset. The clean training set consists of 12.500 images, split to 1.250 images per class. A separate clean validation set is used for model selection and router training, consisting of 500 images (50 per class). All models are trained exclusively on the clean training split.

Synthetic Corruptions. The synthetic corruption benchmark consists of 500 images per corruption type (Gaussian noise and pixelation), evaluated across five severity levels. The dataset covers the same 10 object classes as the clean validation set. These perturbations modify low-level image statistics while preserving semantic content.

Real-World OOD Dataset. We evaluate models on a student-collected dataset containing approximately 300 images per student (30 per class), captured in outdoor or atypical environments using different devices. This dataset introduces shifts in background, lighting, viewpoint, and acquisition conditions.

All splits are strictly separated. No target-domain adaptation or corruption data is used during training.

4.2 Training Protocol

All backbone models are initialized with ImageNet-pretrained weights and fine-tuned on the clean training set. Optimization uses stochastic gradient descent (SGD) with momentum 0.9 and weight decay 10^{-4}. Training runs for up to 30 epochs with early stopping based on best validation accuracy (patience = 5 epochs). Backbone-specific hyperparameter optimization is described in the respective subsections.

For the **homogeneous ensemble**, a two-stage optimization strategy is employed.

In Phase 1, each Resnet is trained independently. Learning rate is optimized via Bayesian optimization over a log-uniform range $[10^{-5}, 10^{-1}]$, while batch size is selected from predefined candidate values. Early termination is implemented using Hyperband. All runs use cosine learning rate scheduling, weight decay 10^{-4}, label smoothing (0.1), and a fixed random seed (42).

In Phase 2, the learning rate search range is narrowed around the best configuration identified in Phase 1, and batch size candidates are restricted to the two best-performing values. Bayesian optimization with Hyperband early stopping is again employed.

The final homogeneous ensemble consists of the best-performing 18-layer, 34-layer and 50-layer ResNet models obtained after Phase 2. For a given input x, each model produces a class probability distribution $p_m(y \mid x)$ via softmax. The ensemble prediction is computed through uniform probability averaging,

$$p_{\text{ens}}(y \mid x) = \frac{1}{M} \sum_{m=1}^{M} p_m(y \mid x),$$

where $M = 3$ denotes the number of ensemble members. The final prediction corresponds to $\arg\max_y p_{\text{ens}}(y \mid x)$. No weighting or routing mechanism is applied in this setup in order to isolate the effect of architectural depth and augmentation-induced diversity.

For the **heterogeneous ensemble** a single, identical data augmentation pipeline is applied to all backbones to ensure fair comparison. Augmentations include RandomResizedCrop, horizontal flipping, and ColorJitter. Keeping the augmentation strategy fixed helps isolate architectural inductive bias as the primary source of diversity in the heterogeneous ensemble.

For the mixture-of-experts setup, backbone weights are frozen. The routing network is trained using cross-entropy on an 80/20 split of validation data to prevent evaluation leakage. No corruption or phone-captured data is used during training.

4.3 Evaluation Metrics

We evaluate all models using classification accuracy, calibration metrics, and oracle analysis. While standard accuracy measures realized performance, oracle accuracy quantifies the theoretical upper bound achievable by the ensemble.

Accuracy. We report overall classification accuracy on clean validation data, synthetic corruption benchmarks, and the phone-captured OOD dataset. For corruption experiments, performance is analyzed across severity levels to assess degradation under increasing shift.

Expected Calibration Error (ECE). Calibration quality is measured using ECE (Guo et al., 2017) computed on the ensemble's final prediction. For each sample, expert probabilities are first combined (via uniform averaging or routing weights), and the maximum predicted probability of the aggregated distribution is taken as the ensemble confidence. Predictions are grouped into 15 equally spaced confidence bins, and ECE is computed as the weighted average of the absolute difference between mean confidence and empirical accuracy within each bin. Lower ECE indicates better alignment between predicted probabilities and observed correctness.

Oracle Accuracy. To assess the potential benefit of expert diversity, we compute oracle accuracy. For each input sample, the oracle selects the expert that predicts the correct class if at least one expert is correct. Formally, oracle accuracy is defined as:

$$\text{Oracle} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\Big(\exists k \text{ s.t. } \hat{y}_k(\mathbf{x}_i) = y_i\Big),$$

where $\hat{y}_k(\mathbf{x}_i)$ denotes the predicted class of expert k for sample $\mathbf{x}_i$.

Oracle accuracy therefore represents the maximum achievable performance under perfect expert selection. The gap between ensemble accuracy and oracle accuracy reflects the extent of complementary expertise within the ensemble. A small gap indicates correlated errors, whereas a large gap suggests substantial specialization potential.

In our analysis (see Section 5), oracle accuracy is used to quantify whether architectural heterogeneity creates exploitable diversity and whether the mixture-of-experts router successfully approaches this theoretical upper bound.

5 Results

5.1 Clean Validation Performance

As shown in Table 2, on the clean validation set, the **homogeneous ensemble** achieves an overall accuracy of 89.20%, compared to 88.20% (18-layer ResNet), 86.00% (34-layer ResNet), and 89.40% (50-layer ResNet). Thus, the ensemble performs comparably to the strongest individual backbone but does not surpass 50-layer ResNet in overall accuracy. Per-class analysis reveals that the ensemble matches or improves upon individual models in several categories (e.g., soup_bowl at 100.0%) while remaining close to the best single-model performance in others. Oracle accuracy reaches 93.60%, indicating additional potential gains if expert selection were optimal. The gap between ensemble and oracle accuracy amounts to 4.4 percentage points. The Expected Calibration Error (ECE) is 0.1458. The average disagreement between ensemble members is 0.0187, corresponding to an agreement ratio of 93.6%.

The **heterogeneous ensemble** achieves strong accuracy (see Table 3) on the clean dataset. Oracle accuracy reaches 96.20%, exceeding the realized ensemble performance. The moderate oracle gap indicates the presence of complementary expertise among backbones. However, uniform averaging already captures most of the attainable performance. Calibration analysis shows that ensemble-level ECE remains non-negligible, indicating that improved accuracy does not automatically imply improved probability calibration.

5.2 Corruption Set

Performance under synthetic corruption is evaluated across five severity levels for Gaussian noise and pixelation.

Figures 1 and 2 show accuracy as corruption severity increases for Gaussian noise and pixelation.

At low corruption levels, both ensembles and the strongest single models perform similarly. Differences are small and remain within a narrow range.

As severity increases, performance begins to diverge. Under Gaussian noise, the homogeneous ResNet ensemble follows the degradation pattern of the single ResNet-50 and exhibits a pronounced drop at higher severity levels. In contrast, the heterogeneous ensemble shows a flatter decline and maintains higher accuracy under severe corruption. The difference becomes most visible at severity levels 4 and 5.

A similar pattern is observed for pixelation. At low levels, both ensembles behave comparably. With increasing distortion, however, the homogeneous ensemble degrades more strongly, while the heterogeneous ensemble remains more stable.

Across both corruption types, the main signal lies in the slope of degradation rather than in absolute accuracy at low severity. Architectural diversity reduces sensitivity to increasing corruption strength, whereas homogeneous ResNet models tend to fail in a more correlated manner under severe shift.

5.3 Data Phone Set / Domain Shift

On the OOD dataset, shown in Table 4, the **homogeneous ensemble** achieves an overall accuracy of 55.8%, compared to 51.1% (18-layer ResNet), 51.6% (34-layer ResNet), and 56.3% (50-layer ResNet). As on the clean validation set, the ensemble performs comparably to the strongest single backbone but does not exceed 50-layer ResNet in overall accuracy. Per-class performance varies substantially across categories, with the ensemble achieving its highest accuracy on *wooden_spoon* (81.2%) and lowest on *soup_bowl* (34.3%). Oracle accuracy reaches 66.75%, resulting in a gap of 10.95 percentage points between ensemble and oracle performance. The Expected Calibration Error (ECE) is 0.0502. The average disagreement between ensemble members increases to 0.0289, corresponding to an agreement ratio of 78.7%.

Performance of the **heterogeneous ensemble** on the phone-captured dataset (see Table 5) exhibits a substantial drop relative to clean validation results, reflecting the severity of real-world distribution shift. Introducing architectural diversity improves realized ensemble accuracy compared to homogeneous CNN aggregation. More importantly, oracle accuracy increases to 72.49%, substantially exceeding realized ensemble performance. The magnitude of this oracle gap is notably larger than on clean data. This indicates that under real-world shift, different experts succeed on different subsets of the data. In other words, complementary expertise becomes more pronounced as the distribution diverges from training conditions. Despite this increased diversity, uniform averaging fails to recover oracle performance. The ensemble remains significantly below its theoretical upper bound. This suggests that the heterogeneous ensemble is not capacity-limited, but limited by its aggregation strategy at inference time.

The presence of substantial unrealized oracle potential motivates the introduction of an adaptive routing mechanism, which aims to exploit input-dependent specialization among experts.

5.4 Mixture-of-Experts Routing

We evaluate whether adaptive expert weighting can reduce the oracle gap observed on the phone-captured dataset. Figure 3 compares three configurations: (i) uniform averaging of the heterogeneous ensemble, (ii) the strongest single backbone (ViT-B/16), and (iii) the learned mixture-of-experts (MoE) router.

Uniform averaging treats all experts equally, regardless of input characteristics. Under real-world shift, this strategy underperforms the strongest single model, indicating that naive aggregation does not fully exploit architectural diversity. Replacing uniform averaging with a learned routing mechanism improves accuracy by approximately 2.4 percentage points relative to simple averaging. This gain is achieved without increasing model capacity or retraining the backbone networks. The improvement therefore reflects more effective utilization of existing expert diversity.

Despite this improvement, routed performance remains substantially below oracle accuracy. A significant portion of the ensemble's theoretical upper bound remains unrealized.

Routing Behavior Analysis. To understand how the router achieves its improvement, we analyze the distribution of learned gating weights. Rather than producing a balanced mixture of experts, the router assigns dominant probability mass to a single expert (see Figure 4). Approximately 85% of the routing weight is allocated to the ViT backbone, with the remaining CNN experts contributing minimally. This behavior indicates that the routing mechanism does not learn fine-grained, input-dependent specialization across experts. As shown in Figure 5, 60.1% of samples are classified correctly by the chosen expert. Missed oracle cases account for 12.4% of samples, representing the only region where improved routing could increase performance without modifying the ensemble. In contrast, 27.5% of samples correspond to hard failures, where no expert predicts correctly.

Class-level Routing Errors. Furthermore, we examine missed oracle cases in which the router selects the ViT backbone despite another expert predicting correctly. Figure 6 shows that these missed opportunities are not uniformly distributed across classes. They are concentrated in specific categories, including *mouse*, *notebook*, and *teapot*. This class-dependent pattern indicates that routing errors are structured rather than randomly distributed across the label space.

ViT Failure Analysis. To analyze routing blind spots, we compare image characteristics between samples where the ViT backbone predicts correctly and samples where it fails. Figure 7 shows that ViT failure cases exhibit systematically higher blur levels. On average, failure images are approximately 32% blurrier than successful predictions. Failure cases also exhibit lower contrast. Importantly, many of these blurry samples are correctly classified by CNN-based experts. For example, DenseNet-121 and EfficientNet-B0 jointly recover a substantial fraction of ViT failures on the *mouse* class. These observations indicate that routing errors are not caused by a lack of alternative experts. Instead, viable CNN predictions exist in regimes where the ViT backbone degrades under reduced image quality.

6 Discussion

6.1 Central Observations

Our findings highlight three central observations regarding ensemble robustness under distribution shift.

First, ensemble improvements depend on diversity among individual model errors (Dietterich, 2000). Uniform averaging provides limited gains when model errors are highly correlated. The oracle analysis demonstrates that robustness benefits depend on the presence of disagreement under shift. When experts succeed on different subsets of samples, ensemble potential increases; when they fail together, aggregation offers little advantage.

Second, architectural diversity is a primary driver of such complementary failure modes. Under real-world shift, heterogeneous ensembles exhibit larger oracle gaps than homogeneous CNN ensembles. This suggests that distinct inductive biases degrade differently under distribution changes. As a result, heterogeneous ensembles maintain robustness in regimes where homogeneous ensembles experience correlated collapse.

Third, our mixture-of-experts experiments reveal a limitation of naive routing strategies. Although routing improves performance relative to uniform averaging, the learned gating mechanism concentrates the majority of its weight on a single dominant expert. The breakdown of routing decisions shows that only a limited fraction of errors correspond to oracle cases, while a substantial portion arises from hard failures where no expert is correct.

Taken together, these results suggest that architectural diversity alone is insufficient to guarantee effective specialization. In the absence of explicit incentives, routing mechanisms tend to prioritize globally strong experts rather than conditionally coordinating complementary ones. Consequently, ensemble robustness under domain shift is constrained both by the diversity of available experts and by the ability of the aggregating mechanisms to exploit that diversity.

Future work may investigate routing objectives that explicitly encourage expert specialization or incorporate input-level uncertainty cues. Alternatively, increasing expert diversity through data-centric or representation-level strategies may reduce the proportion of hard failures observed under real-world shift.

6.2 Limitations

Several limitations contextualize our findings.

First, backbone representations are not adapted to target domains. All experts are trained exclusively on the clean in-domain dataset. Consequently, observed robustness gains arise purely from aggregation strategies rather than improved feature learning. It remains unclear whether joint adaptation or domain-aware representation learning would reduce the proportion of hard failures observed under real-world shift.

Second, the routing network is trained on a relatively small validation split. The limited training data may constrain the router's ability to learn fine-grained specialization and likely contributes to the observed concentration of routing weights on a single dominant expert. A larger or more diverse routing dataset may enable more conditional coordination across experts.

Third, the synthetic corruption benchmark (Gaussian noise and pixelation) captures low-level distribution shifts but does not model semantic or contextual variation. Real-world domain shift often involves changes in background, object context, and acquisition conditions that extend beyond pixel-level distortions.

Taken together, these limitations indicate that both representation quality and expert diversity jointly constrain ensemble robustness under domain shift.

6.3 Conclusion

This work examined ensemble robustness under distribution shift and provides three central insights. First, robustness gains are not a function of model count but of failure diversity. Ensembles improve only when experts make complementary errors rather than correlated ones.

Second, architectural heterogeneity primarily benefits robustness under shift, not clean accuracy. Distinct inductive biases degrade differently, creating complementary behavior that can be exploited under corruption and real-world variation.

Third, routing mechanisms do not automatically induce specialization. In our setting, the mixture-of-experts model behaves largely as expert selection rather than coordinated collaboration. Without explicit incentives, gating mechanisms favor globally strong experts instead of conditionally combining complementary ones.

Together, these findings suggest that ensemble robustness emerges less from stronger individual models and more from structured diversity and effective aggregation. Designing ensembles for robustness therefore requires intentional diversity engineering and routing objectives that promote specialization rather than dominance.

Author Contributions

Sebastian implemented and evaluated the homogeneous ensemble baseline, including training, validation, and corruption robustness analysis. Sebastian wrote the Background section, the homogeneous ensemble subsection of Methods, and contributed to the Experimental Setup and Results sections. Sebastian also authored the Conclusion.

Stephanus was responsible for implementation of the heterogeneous ensemble and the mixture-of-experts routing mechanism. This included backbone selection, routing architecture design, experimental execution, and routing-behavior analysis. Stephanus wrote the Introduction, the heterogeneous ensemble and routing subsections of Methods, the corresponding Results subsections, and the Discussion and Limitations sections.

Both authors jointly designed the experimental setup, reviewed all sections of the report, and contributed to interpretation of results and manuscript revisions.

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Declaration of Authorship

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A Appendix

Table 2: Per-class accuracy (%) on the clean dataset. Columns compare the homogeneous ensemble with its constituent backbones (ResNet-18, ResNet-34, and ResNet-50). The overall row reports aggregate accuracy across classes.

Class	Ensemble	ResNet-18	ResNet-34	ResNet-50
binder	**98.0**	96.0	86.0	**98.0**
coffee_mug	78.0	78.0	76.0	**82.0**
computer_keyboard	**82.0**	**82.0**	78.0	76.0
mouse	78.0	66.0	**82.0**	78.0
notebook	78.0	**86.0**	80.0	82.0
remote_control	94.0	90.0	92.0	**96.0**
soup_bowl	**100.0**	98.0	96.0	98.0
teapot	98.0	**100.0**	96.0	98.0
toilet_tissue	90.0	**92.0**	86.0	88.0
wooden_spoon	96.0	94.0	88.0	**98.0**
Overall	89.2	88.2	86.0	**89.4**

Table 3: Per-class accuracy (%) on the clean dataset. Columns compare the heterogeneous ensemble with its constituent backbones (DenseNet-121, EfficientNet-B0, ResNet-34, and ViT-B/16). The overall row reports aggregate accuracy across classes.

Class	Ensemble	DenseNet121	EfficientNet	ResNet-34	ViT-B/16
binder	**100.0**	94.0	96.0	94.0	**100.0**
coffee_mug	82.0	78.0	76.0	82.0	**84.0**
computer_keyboard	86.0	76.0	76.0	80.0	**88.0**
mouse	80.0	74.0	82.0	72.0	**86.0**
notebook	84.0	82.0	**86.0**	82.0	80.0
remote_control	**96.0**	94.0	**96.0**	88.0	**96.0**
soup_bowl	**98.0**	**98.0**	96.0	**98.0**	94.0
teapot	**100.0**	98.0	**100.0**	96.0	**100.0**
toilet_tissue	**96.0**	92.0	94.0	92.0	94.0
wooden_spoon	**96.0**	**96.0**	94.0	94.0	94.0
Overall	**91.8**	88.2	89.6	87.8	91.6

Table 4: Per-class accuracy (%) on the phone-captured out-of-distribution dataset. Columns compare the homogeneous ensemble with its constituent backbones (ResNet-18, ResNet-34, and ResNet-50). The overall row reports aggregate accuracy across classes.

Class	Ensemble	ResNet-18	ResNet-34	ResNet-50
binder	55.3	51.5	45.6	**58.4**
coffee_mug	48.5	42.7	39.7	**54.9**
computer_keyboard	42.6	39.6	36.8	**44.7**
mouse	52.9	46.7	**56.3**	48.7
notebook	**66.0**	62.5	60.0	64.9
remote_control	62.7	52.9	58.8	**64.0**
soup_bowl	34.3	30.4	30.8	**35.3**
teapot	37.1	32.7	**40.0**	35.8
toilet_tissue	77.9	**78.8**	70.3	75.1
wooden_spoon	81.2	74.1	78.2	**81.4**
Overall	55.8	51.1	51.6	**56.3**

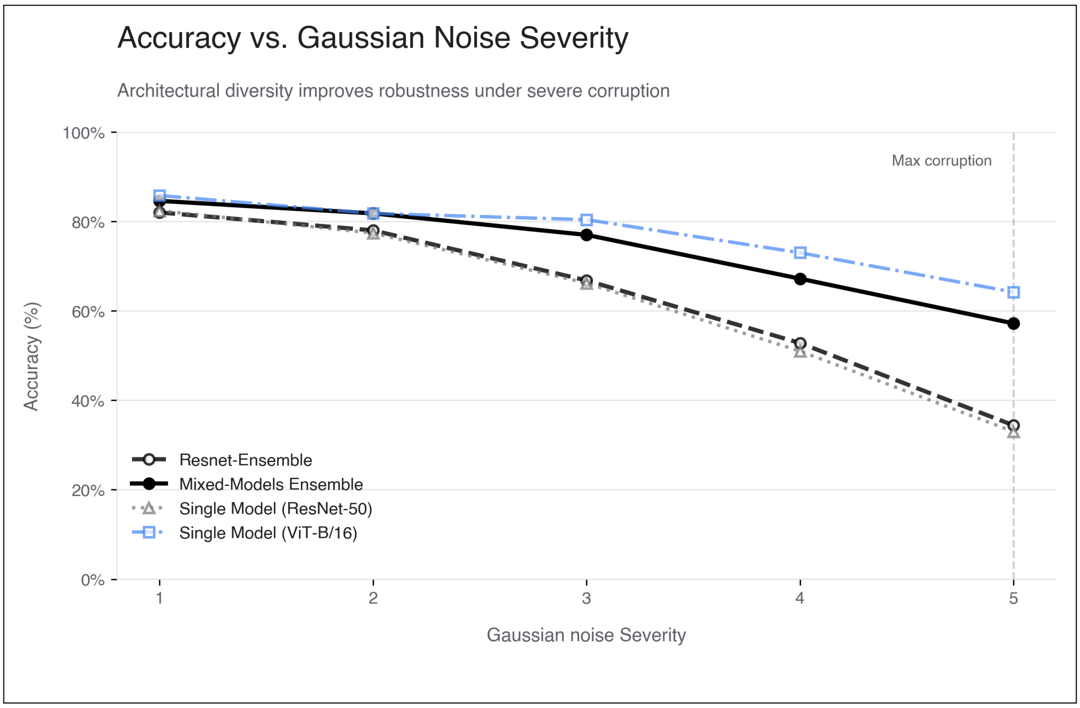

Figure 1: Classification accuracy under Gaussian noise corruption across five severity levels. Results are shown for a homogeneous ResNet ensemble, the heterogeneous mixed-model ensemble, and the strongest single backbones (ResNet-50 and ViT-B/16). Evaluation is performed on the synthetic corruption benchmark.

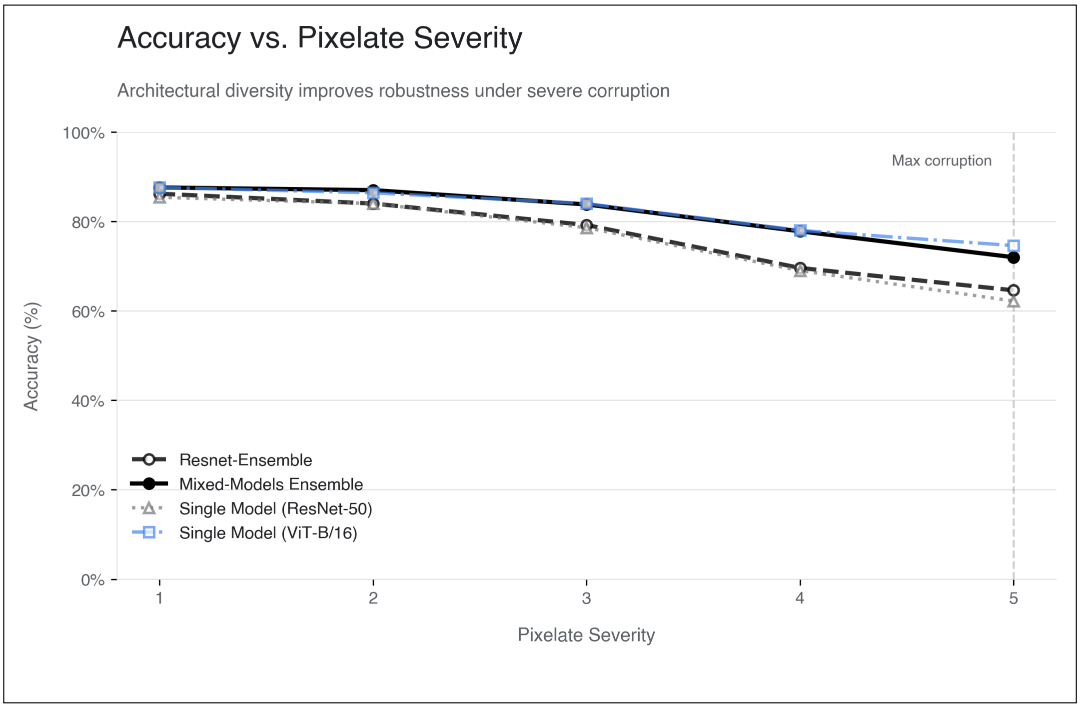

Figure 2: Classification accuracy under pixelation corruption across five severity levels. Results are shown for a homogeneous ResNet ensemble, the heterogeneous mixed-model ensemble, and the strongest single backbones (ResNet-50 and ViT-B/16). All evaluations are conducted on the synthetic corruption benchmark.

Table 5: Per-class accuracy (%) on the phone-captured out-of-distribution dataset. Columns compare the heterogeneous ensemble with its constituent backbones (DenseNet-121, EfficientNet-B0, ResNet-34, and ViT-B/16). The overall row reports aggregate accuracy across classes.

Class	Ensemble	DenseNet121	EfficientNet	ResNet-34	ViT-B/16
binder	56.9	38.2	51.7	48.5	**68.3**
coffee_mug	49.1	48.5	41.2	38.6	**57.1**
computer_keyboard	44.3	39.3	33.0	37.7	**54.8**
mouse	49.2	46.9	**49.4**	41.9	43.5
notebook	74.0	71.9	**74.5**	62.3	67.0
remote_control	57.5	54.4	55.7	44.3	**60.5**
soup_bowl	**32.5**	29.7	30.6	24.8	32.0
teapot	44.8	38.0	41.7	40.6	**49.2**
toilet_tissue	**86.4**	82.5	74.0	80.6	84.3
wooden_spoon	**86.0**	81.4	84.6	81.2	84.6
Overall	58.0	53.0	53.6	49.9	**60.1**

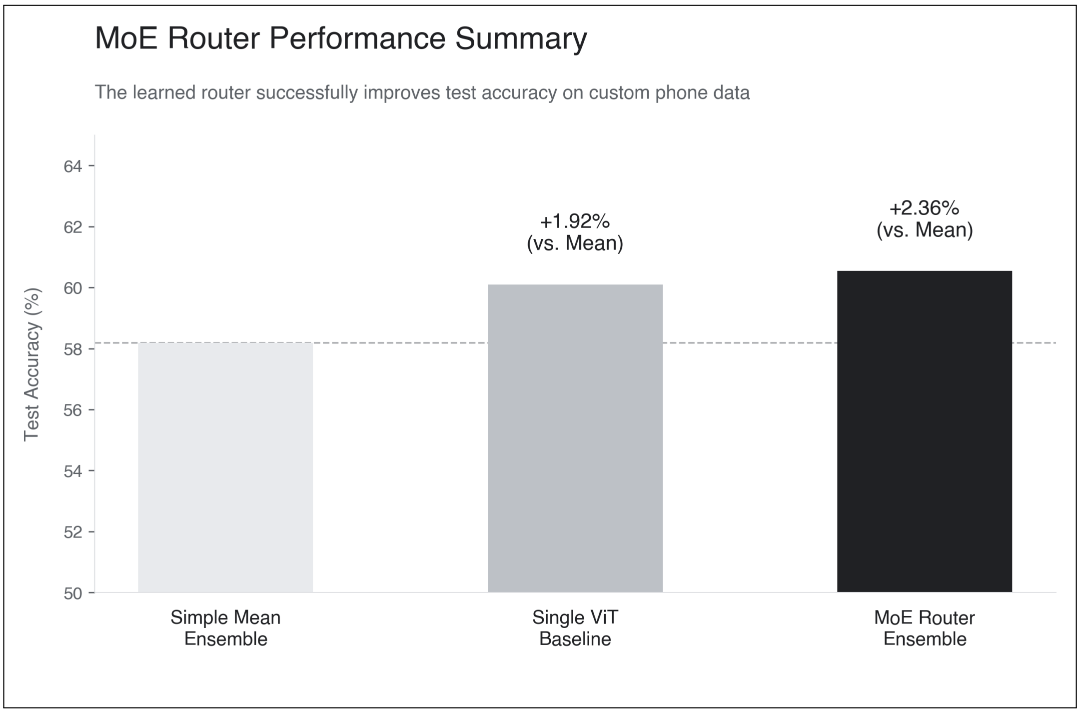

Figure 3: Test accuracy on the phone-captured dataset for uniform averaging (Simple Mean Ensemble), the strongest single backbone (ViT-B/16), and the Mixture-of-Experts (MoE) router. Percentage annotations indicate relative improvement over uniform averaging.

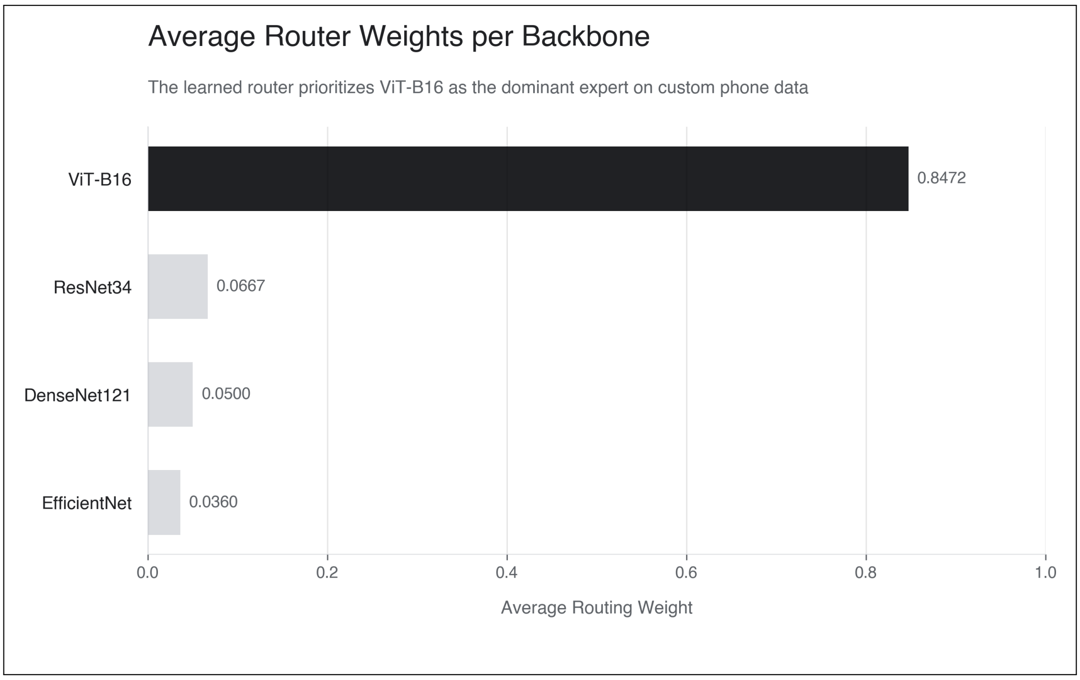

Figure 4: Average routing weight assigned to each backbone by the mixture-of-experts router on the phone-captured dataset. Weights are averaged across all samples. The router assigns the majority of probability mass to the ViT-B/16 backbone.

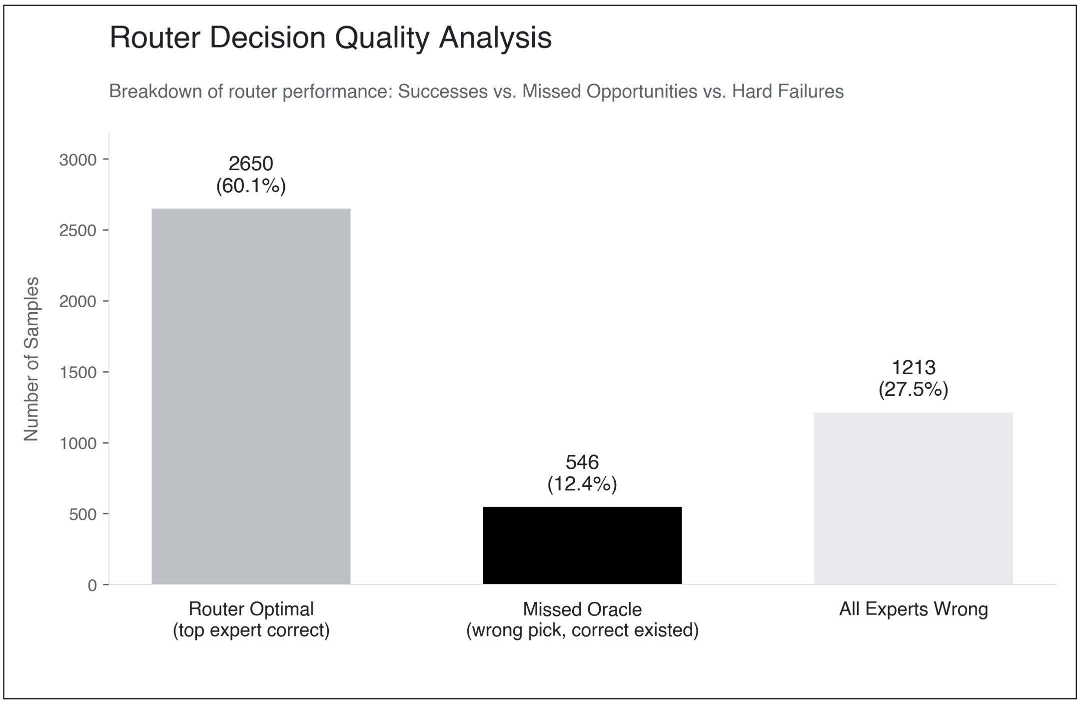

Figure 5: Breakdown of routing outcomes on the phone-captured dataset. Samples are categorized as: (i) router selects a correct expert (“Router Optimal”), (ii) router selects an incorrect expert although another expert predicts correctly (“Missed Oracle”), and (iii) all experts predict incorrectly (“All Experts Wrong”). Percentages indicate the proportion of total evaluation samples in each category.

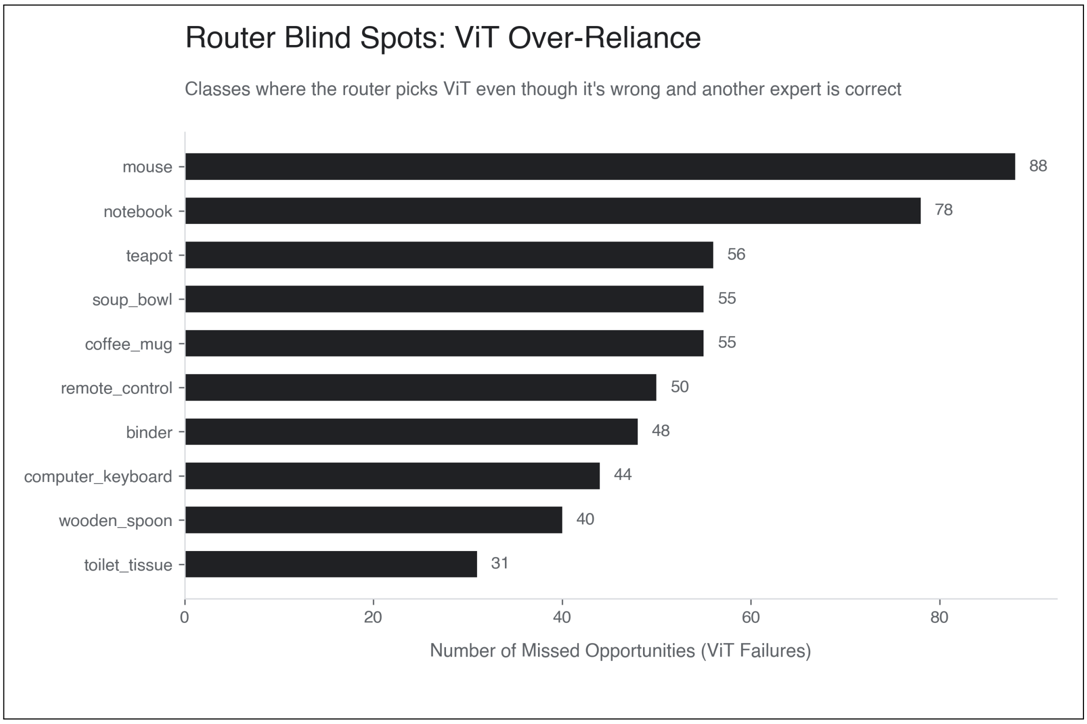

Figure 6: Class-wise distribution of missed oracle cases on the phone-captured dataset. The bars indicate the number of samples per class where the router selects the ViT backbone despite another expert predicting correctly.

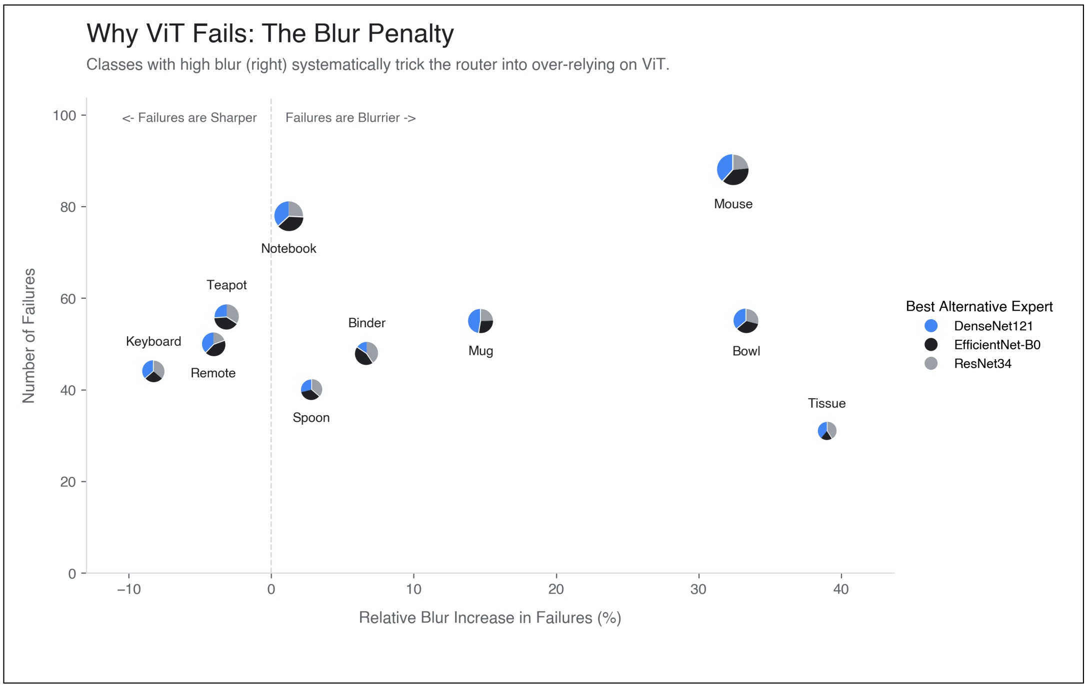

Figure 7: Analysis of ViT failure characteristics on the phone-captured dataset. The x-axis shows the relative increase in blur for failure cases compared to successful predictions, and the y-axis indicates the number of failures per class. Pie charts denote the alternative expert (DenseNet-121, EfficientNet-B0, or ResNet-34) that correctly predicts the label in these cases.